

CS & IT ENGINEERING



Computer Network

Introduction

Lecture No. - 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

Application Layer

Topic

Two Process Communication





Topics to be Covered



Topic

Transport Layer

Topic

Network Layer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Presentation Layer



⇒ Big Endian format supported by N/w

Presentation layer functions :-

1. Code Conversion
2. Encryption and Decryption
3. Compression and Decompression

(optional)





Topic : Code Conversion



→ Convert data into network standard code, before transmission
[Network presentable form]



ASCII : American Standard Code for Information Interchange

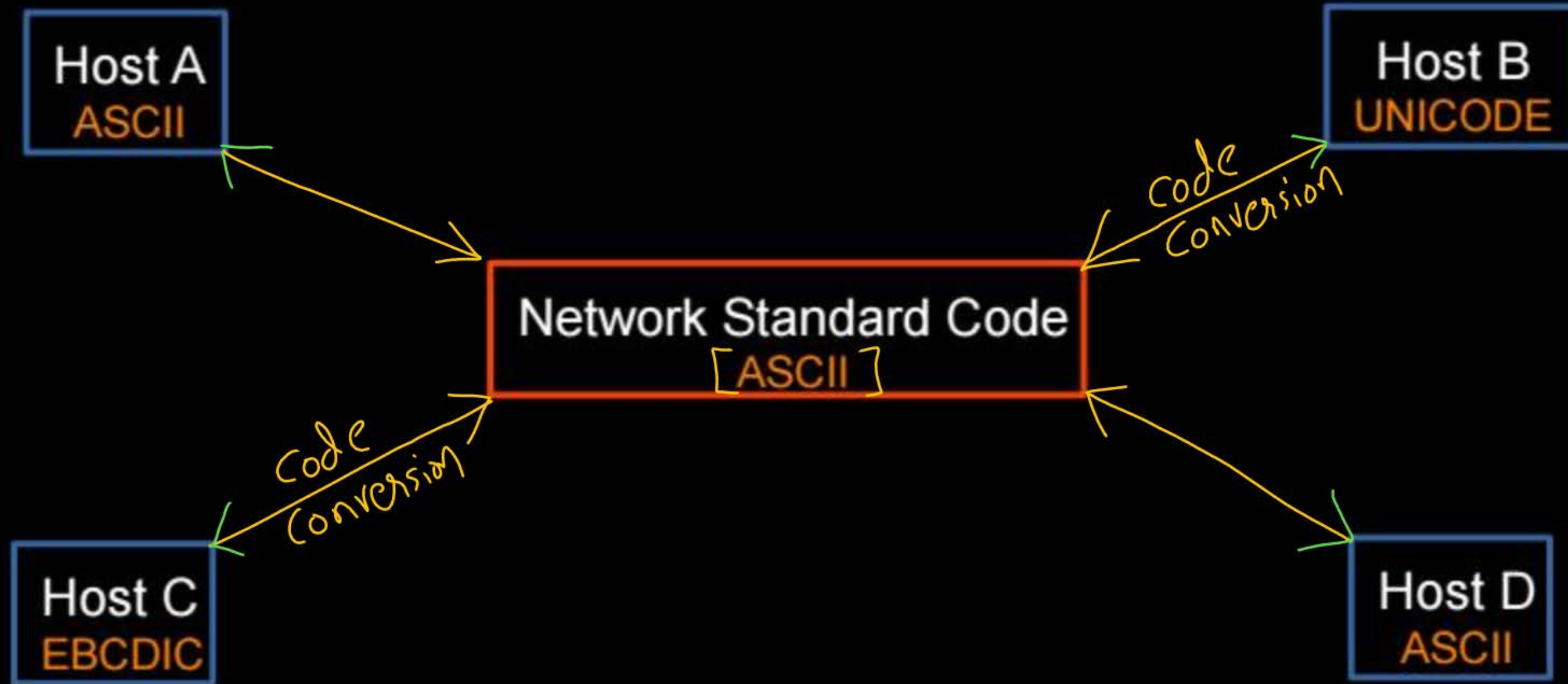
EBCDIC : Extended Binary Coded Decimal Interchange Code

UNICODE : Universal Coded Character Set

→ A = 65



Topic : Code Conversion





Topic : Session Layer

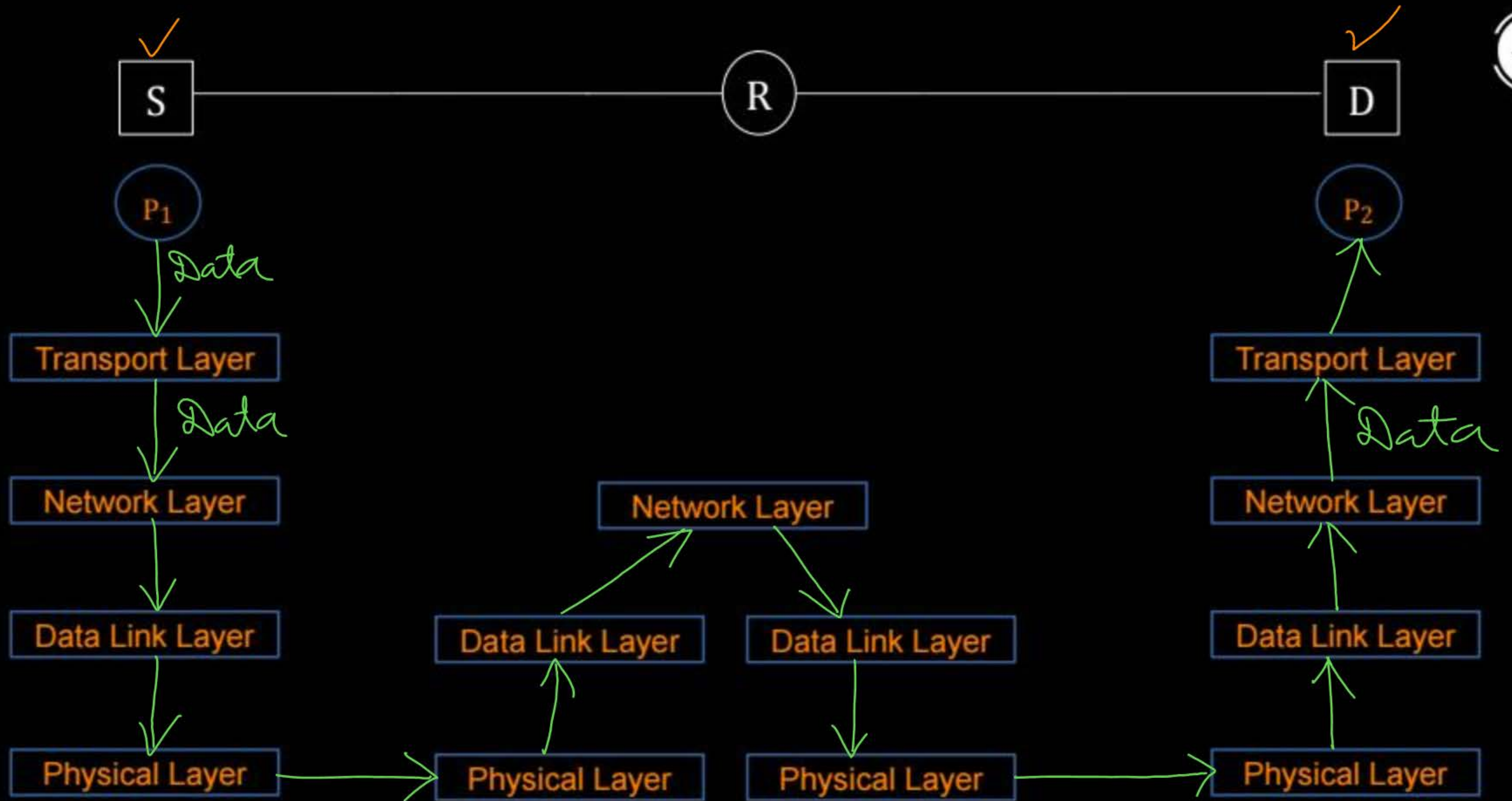


Session layer functions :-

1. Session Establishment
2. Dialog Management
3. Authentication
4. Authorization

#Q. Assume that source and destination hosts are connected through one intermediate router. Determine how many times each packet has to visit the data link layer during a transmission from source to destination hosts?

Ans = 4



#Q. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D ?



[GATE-2013, 1-Mark]

11T-B
[H.W.]

- (A) Network layer – 4 times and Data link layer – 4 times
- (B) Network layer – 4 times and Data link layer – 3 times
- (C) Network layer – 4 times and Data link layer – 6 times
- (D) Network layer – 2 times and Data link layer – 6 times



Topic : Transport Layer



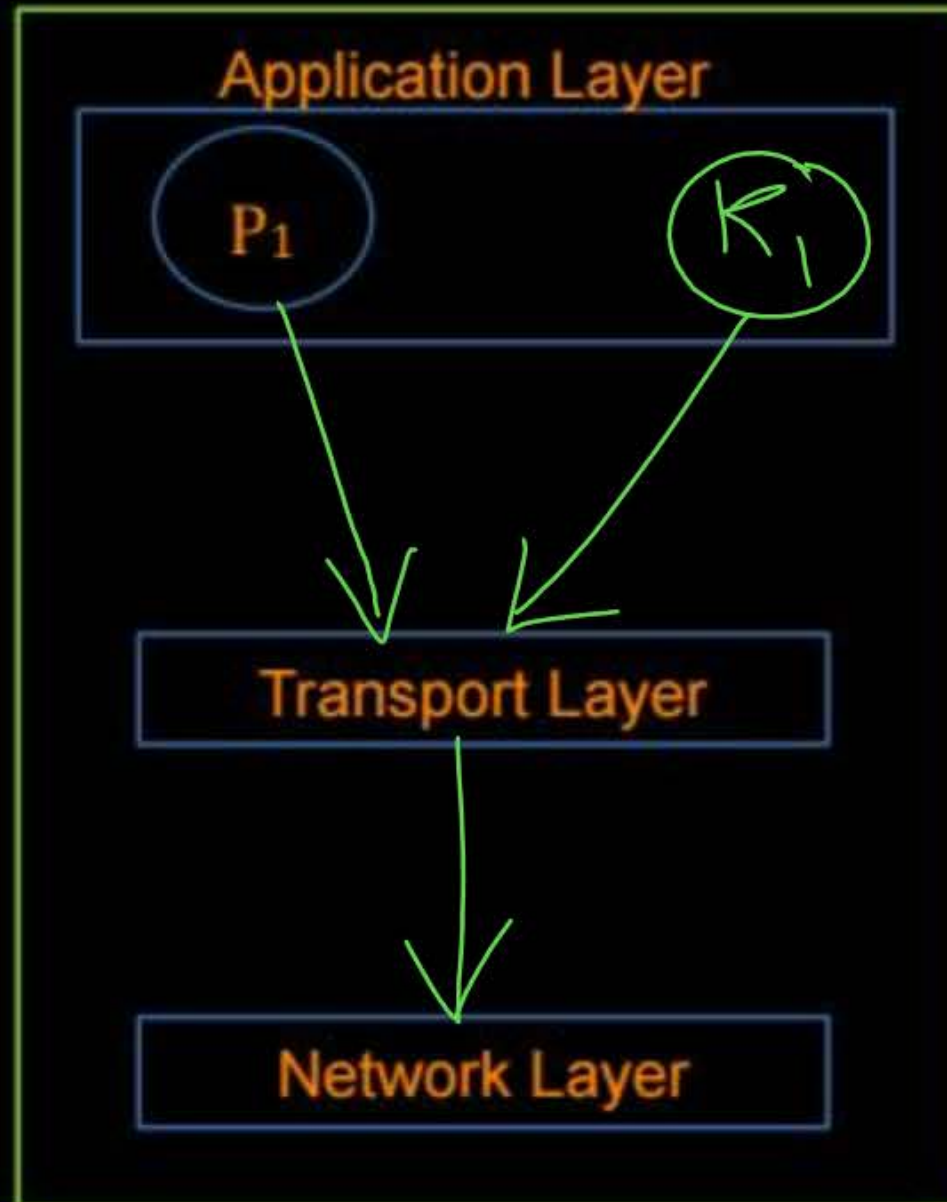
- Provide logical communication between application processes
(Processes running on different machine)
- Provide process-to-process (end-to-end) connectivity



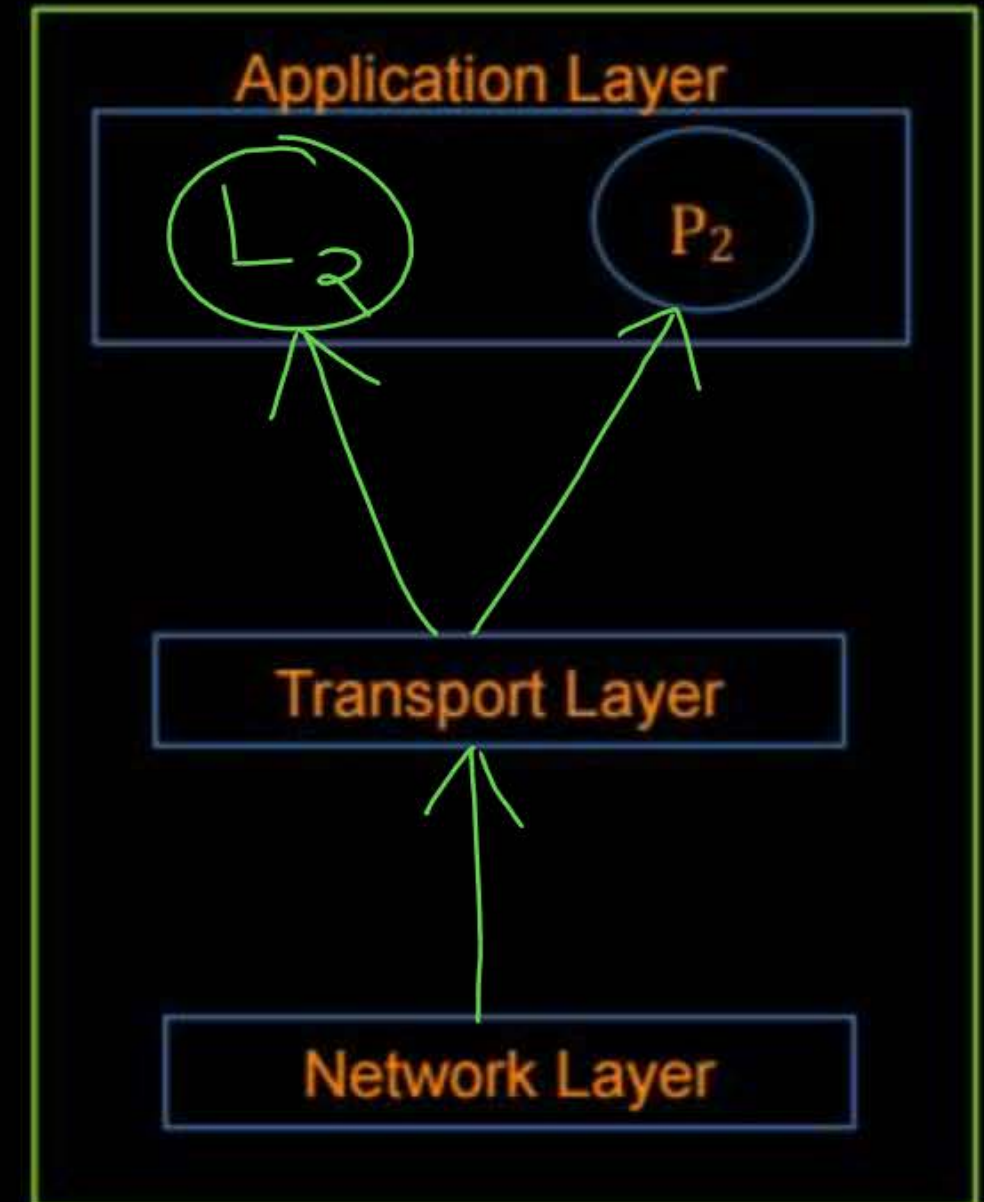
Topic : Process-to-process connectivity



Source Host



Destination Host



#Q. Which of the following functionality must be implemented by a transport protocol over and above the network protocol?

[GATE-2003]

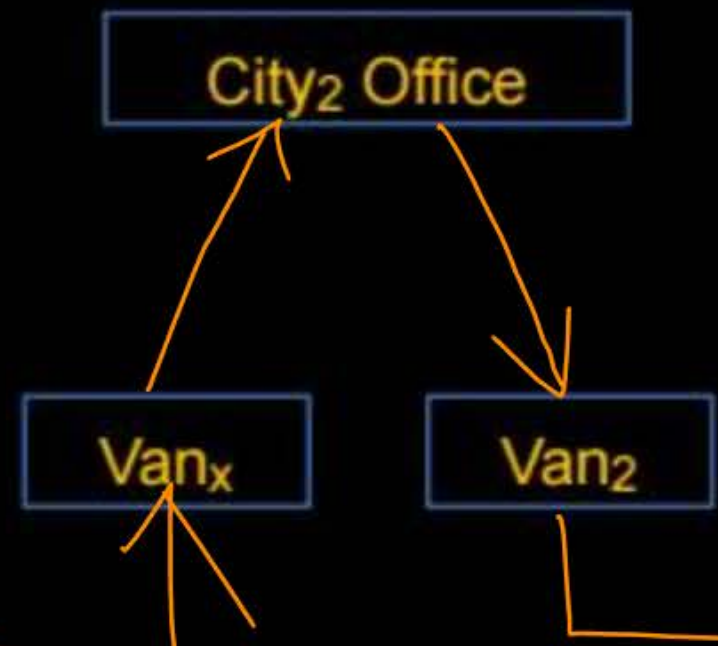
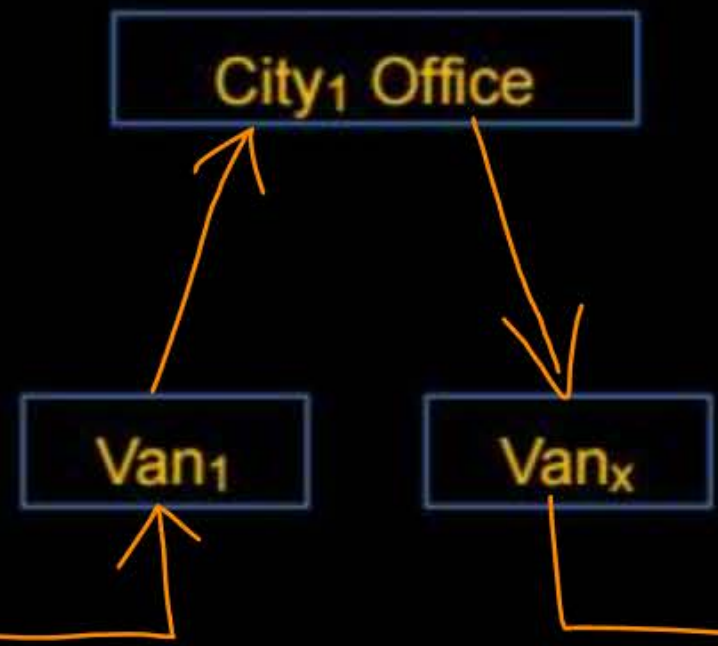
IIT-M

- ☒ (A) Recovery from packet losses
- ☒ (B) Detection of duplicate packets
- ☒ (C) Packet delivery in the correct order
- ☐ (D) End-to-end connectivity

(Optional)

Ans: D

(K_3) (L_3)
City₃





Topic : Transport Layer



→ Multiplexing & Demultiplexing ✓

[Demultiplexing on the basis of Destination Port Number] →



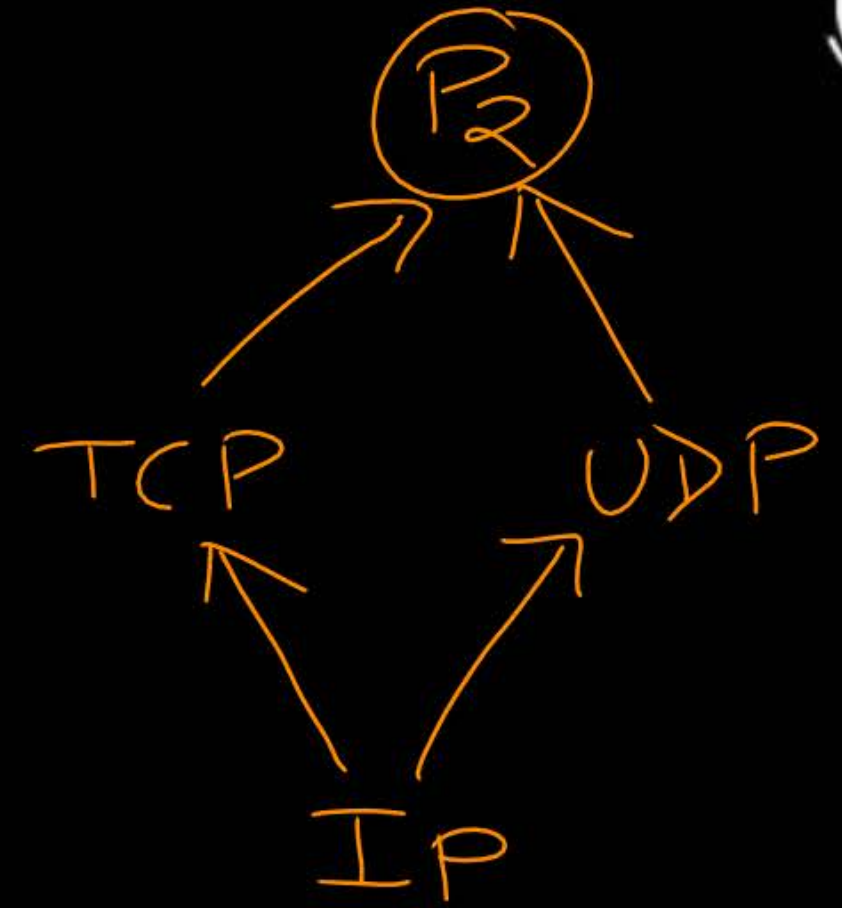
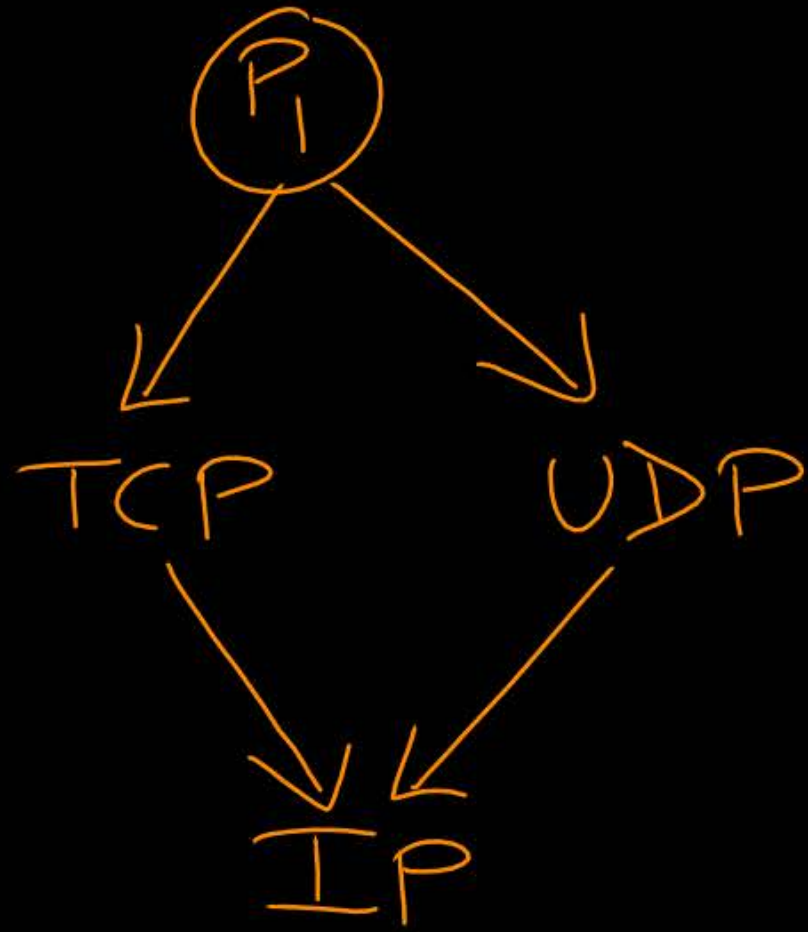
Topic : Transport Layer

Two Transport Layer Protocols :

1. UDP : User Datagram Protocol $\Rightarrow$ Basic Protocol
2. TCP : Transmission Control Protocol

$[TCP = UDP + \text{Extra Services}]$

$\rightarrow$ UDP is Light weight Protocol
 $[it \text{ is providing very minimal services}]$



→ UDP provide faster communication,
in compare to TCP.

→ TCP provide Reliable communication



Topic : Transport Layer

Application Layer Protocol	Transport Layer Protocol
<u>DNS</u>	<u>UDP</u> (Default) ✓ <u>TCP</u> (Conditional)
HTTP/1 HTTP/2	TCP
HTTP/3	UDP
FTP	TCP
SMTP	TCP

#Q. Which one of the following uses UDP as the transport protocol?

[GATE-2007]

IIT-K

- ~~(A) HTTP~~ 1 & 2 → TCP
- ~~(B) Telnet~~ → TCP
- ☒ (C) DNS → UDP (default)
- ~~(D) SMTP~~ → TCP

Ans: C

#Q. Which of the following transport layer protocols is used to support electronic mail?

[GATE-2012, 1-Mark]

Email: SMTP

- ☒ (A) SMTP APP. Protocol
- ☒ (B) IP Network Protocol
- ☐ (C) TCP
- ☒ (D) UDP

Ans: C



#Q. Which of the following transport layer protocols is used to support electronic mail?

[GATE-2012, , 1-Mark]

IIT-D

- (A) SMTP
- (B) IP
- (C) TCP
- (D) UDP

Ans : (C) TCP

E-mail uses SMTP as application layer protocol.
SMTP uses TCP as transport layer protocol.



2 mins Summary



Topic

Transport Layer

Topic

~~Network Layer~~



THANK - YOU

